Checkpoint 6

Data 6560 Sport Analytics
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Risk vs. Reward: Analyzing Aggressive vs. Conservative Decision-Making in Golf

Question and Outcome

- The primary question at hand is 'Does a golfer's aggressiveness, measured by ball striking aggressiveness (bs_aggr) predict overall performance?
- This question is evaluated using SG_total as the outcome variable
- Two samples are used, all of 2022 PGA events, and the 2022 Masters Tournament only

Data Used

- Row = one player-tournament round
- Masters sample and full 2022 sample processed separately
- Variables: sg_total, sg_t2g, sg_putt, sg_app, sg_ott, bs_aggr, aggr_split
- Filters: rows missing key SG variables removed

Method Choice

The method for this analysis was a group mean comparison + OLS regression analysis. This was done to quantify differences in performance by aggressiveness and isolate partial effects. The baseline was mean SG_Total by aggressiveness group, and the improved regression included bs_aggr and sg_putt.

To carry out the regression, python code was used, run through Google Colab. The code is listed in the GitHub repository under Ball Striking Analysis.py

Analysis Spec

Outcome: SG_total, Predictors: bs_aggr and sg_putt, Samples: Masters-only 2022 and Full 2022 files, Expected Direction: More aggressive shows higher sg_total

```
# Convert 'bs_aggr' to numeric, coercing errors, then drop NaNs
df['bs_aggr'] = pd.to_numeric(df['bs_aggr'], errors='coerce')
df.dropna(subset=['bs_aggr'], inplace=True)

# create groups by bs_aggr quartiles
df['vol_q'] = pd.qcut(df['bs_aggr'], 4, labels=['Q1','Q2','Q3','Q4'])
q25 = df['bs_aggr'].quantile(0.25)
q75 = df['bs_aggr'].quantile(0.75)
def vol_label(x):
    if x <= q25: return 'Low'
    if x >= q75: return 'High'
    return 'Mid'
df['aggr_group'] = df['bs_aggr'].apply(vol_label)

# group summary
group_stats = df.groupby('aggr_group').agg(
    n=('player','count'),
    mean_sg_total=('sg_total','mean'),
    median_sg_total=('sg_total','median'),
    mean_score_par=('score_par_diff','mean')
```

Results

High aggressiveness avg SG_Total = 1.26
Mid = -0.14
Low = -2.22
T-test high vs low: t = 127.27, $p < 0.001$

Correlation Matrix Highlights:

bs_aggr and sg_total have a strong correlation (~0.72)
Aggressive split is the only weak correlation
Putting moderately correlated to performance

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               sg_total   R-squared:                       0.853
Model:                            OLS   Adj. R-squared:                  0.853
Method:                 Least Squares   F-statistic:                 8.479e+04
Date:                Wed, 10 Dec 2025   Prob (F-statistic):               0.00
Time:                        02:15:48   Log-Likelihood:                -33146.
No. Observations:               29180   AIC:                         6.630e+04
Df Residuals:                   29177   BIC:                         6.632e+04
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0363      0.004     -8.149      0.000      -0.045      -0.028
bs_aggr        0.9987      0.003    329.200      0.000       0.993       1.005
sg_putt        1.0063      0.004    255.290      0.000       0.999       1.014
==============================================================================
Omnibus:                     3160.666   Durbin-Watson:                   2.014
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            14046.784
Skew:                          -0.457   Prob(JB):                         0.00
Kurtosis:                       6.274   Cond. No.                         1.52
```

Regression Results:

For the full 2022 pga events, the R^2 was 0.853, bs_aggr coef ~ 1.00 ($p < 0.001$), sg_putt coef ~ 1.01 ($p < 0.001$)

Interpretation: +1 in bs_aggr correlates to +1 sg_total stroke per round

For the 2022 masters only, the R^2 was 0.814, bs_aggr coef ~ 0.94, sg_putt coef ~ 0.95
Interpretation: the trend holds in a snapshot setting at an elite level event

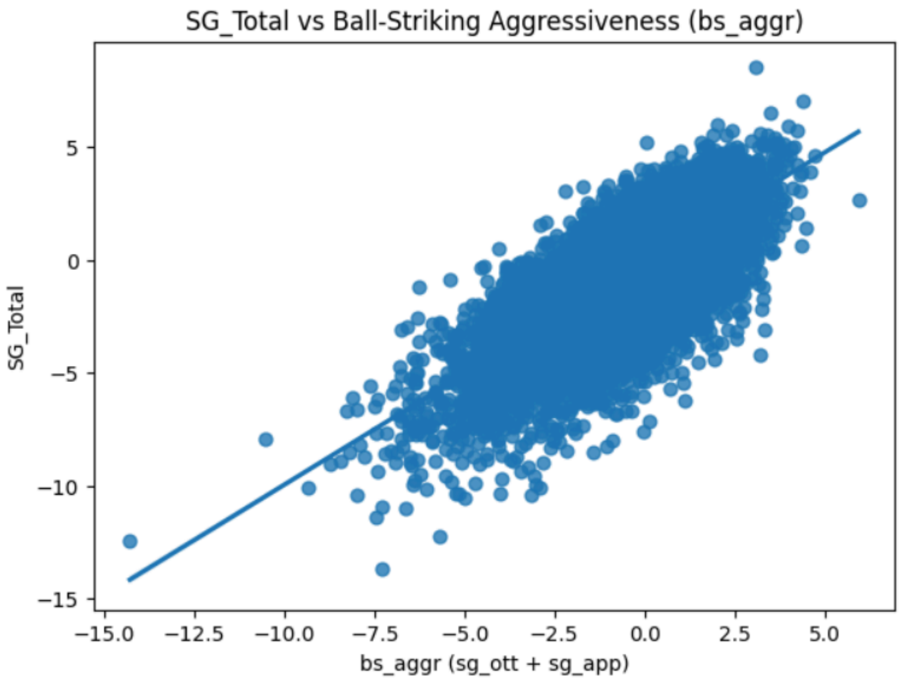

Fig. 1 – Full 2022 season correlation visualization between sg_total and bs_aggr

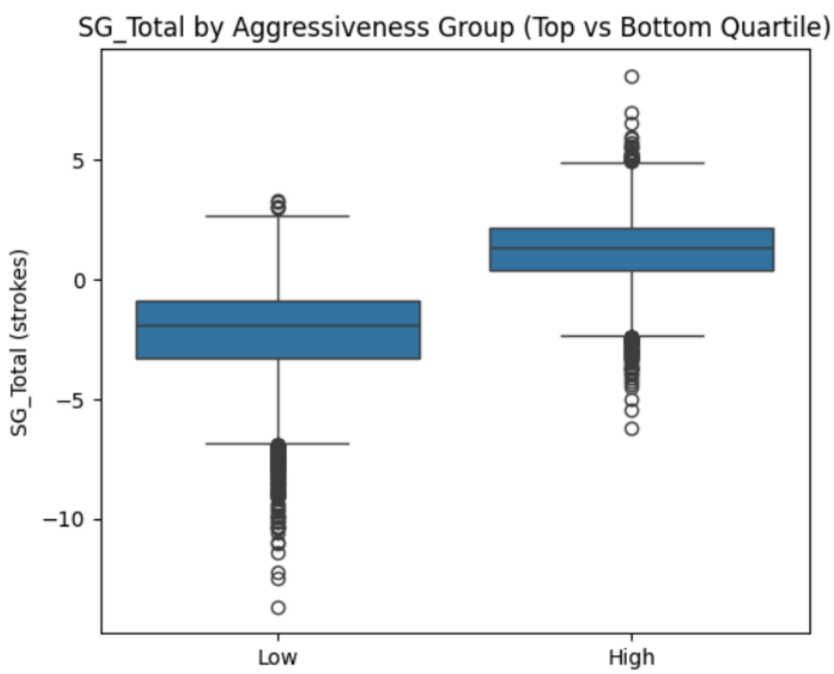

Fig. 2 – Full 2022 season box plot between top and bottom quartile in SG by aggressiveness (bs_aggr)

Checks

There was no leakage as strokes gained were round-level. The results are stable across multiple samples, and the predictors are on consistent scales.

Interpretation

Aggressive players outperform conservative players, putting performance boosts or limits returns to aggressiveness, and the effect seen is consistent across both samples. In the real world, these results are telling golfers to be aggressive with their ball striking as it will lead to improved results against the rest of the field of play. Increasing aggressiveness heavily correlates to strokes gained. So in context, when deciding to go for an aggressive vs conservative play, going for it and accepting the higher risk is backed by the data to potentially score better.

We are starting to see more of an adoption of this idea in recent years with players hitting further/more aggressively off of the tee since it puts them in a better position to make scores under par. The old school idea of ‘consistency is king’ remains partially true, as putting performance still plays a big impact once a player reaches the green, but from tee to green aggressive plays should be considered.

Limitations

The aggressiveness metric is only for one season (2022) and on a round level, not accounting for shot by shot data. Strokes gained also varies by course difficulty, which was not entirely accounted for in this analysis.

Next Steps

The next steps would be to add volatility as a second feature, try some mixed models, and test predictive accuracy for a season outside of 2022.